



HAMBURG RACE PREVIEW

Elite Analysis · July 9 – 12, 2026

USA Triathlon High Performance

Race Week Schedule

Hamburg



July 9 – 12, 2026 • Rathausmarkt / Binnenalster, Hamburg, Germany (CEST, UTC+2)

Fri • July 10	
Familiarization & Briefing	
TBD	Course familiarization — swim, bike, run circuits
TBD	★ Mandatory Elite athlete briefing
TBD	Race package distribution

Sat • July 11	
ELITE SPRINT RACE DAY	
11:30 – 12:45	Elite Men — athletes' lounge check-in + warm-up
12:15 – 13:00	Elite Men — swim warm-up + transition check
13:10	Elite Men — athletes' introduction
13:15	★ ELITE MEN START
~14:06	Elite Men — projected finish
14:15	Elite Men medals ceremony
13:45 – 14:45	Elite Women — athletes' lounge check-in + warm-up
14:15 – 14:55	Elite Women — swim warm-up + transition check
14:50	Elite Women — athletes' introduction
15:00	★ ELITE WOMEN START
~15:55	Elite Women — projected finish
16:05	Elite Women medals ceremony
Evening	Mixed Relay team declaration

Sun • July 12	
Mixed Relay Day	
TBD	Mixed Relay team managers' meeting
17:30	★ MIXED RELAY START
~18:45	Mixed Relay projected finish
Post-finish	Mixed Relay medals ceremony

Times Rathausmarkt / Binnenalster, Hamburg, Germany (CEST, UTC+2). Schedule confirmed from World Triathlon race-info page / athlete guide; subject to organiser updates.

Key Course Differentiators

Hamburg



HAMBURG

WTCS Hamburg is one of the sport's flagship city-center races, set in the heart of Hamburg along the Binnenalster and the historic Rathausmarkt. The sprint format rewards athletes who can handle high-frequency cornering on the bike and can run at the pointy end of the field. Hamburg has hosted elite triathlon since 2008 and consistently delivers fast, tactical racing in front of one of the largest crowds on the WTCS calendar.

COURSE FEATURES

- Sheltered 750 m Binnenalster swim. Typically flat fresh water. Pontoon start and likely a non-wetsuit swim.
- 6 × 3.3 km urban bike (19.8 km). Flat but highly technical course with multiple turns. Large packs historically come together.
- 2 × 5 km flat run through city-center streets. Despite some cobblestone sections this is typically a fast run that comes down to a late surge.
- Elite Men 13:15 / Elite Women 15:00 Saturday July 11. Mixed Relay Sunday July 12.

COURSE AT A GLANCE

Swim:	750 m – 1 lap
Bike:	19.8 km – 6 laps × 3.3 km
Run:	10 km – 2 laps × 5 km
Format:	Sprint Triathlon
Bike Laps:	6
Run Laps:	2

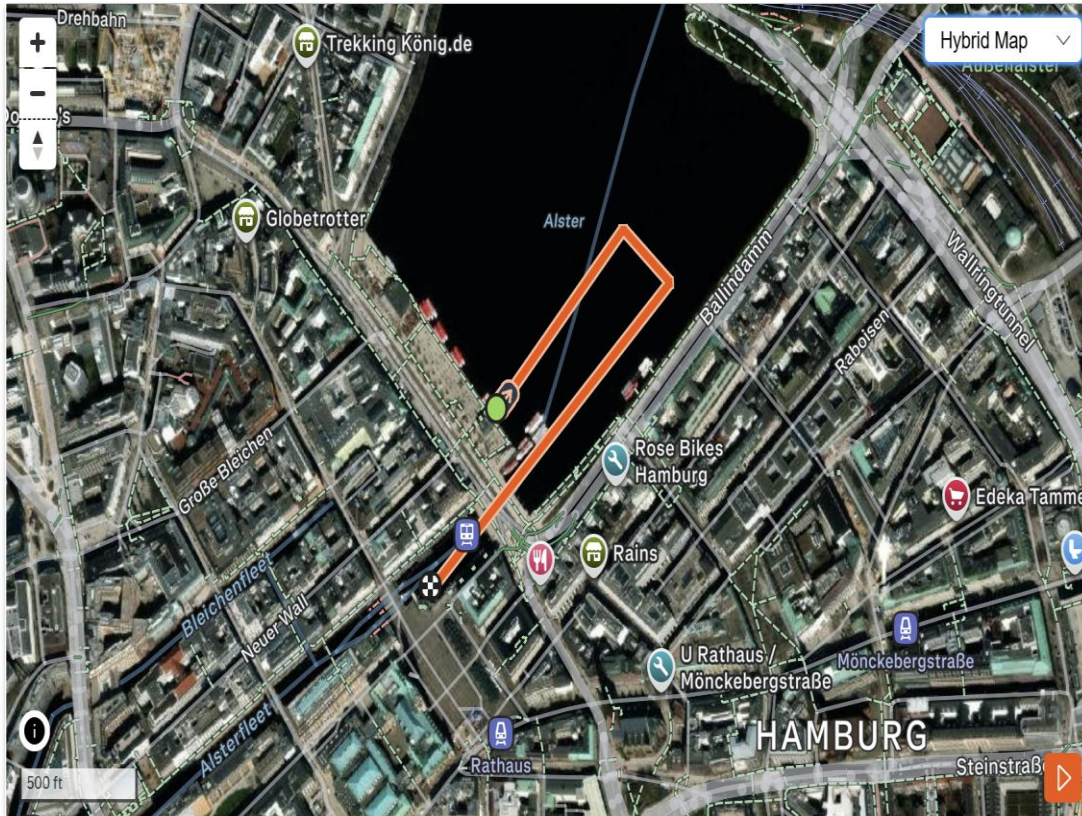
QUICK LINKS

Current Race Info:

[Race Info | 2026 World Triathlon Championship Series Hamburg](#)

Swim Course Profile

Hamburg



Source: World Triathlon race info — 2026 WTCS Hamburg

750 m • single lap

Sheltered urban lake — fresh water, calm conditions

Distance

750 m

Body of Water

River

Start

Pontoon start

Water Temp

18.0–22.0 °C

Wetsuit

Likely non-wetsuit

Lap Count

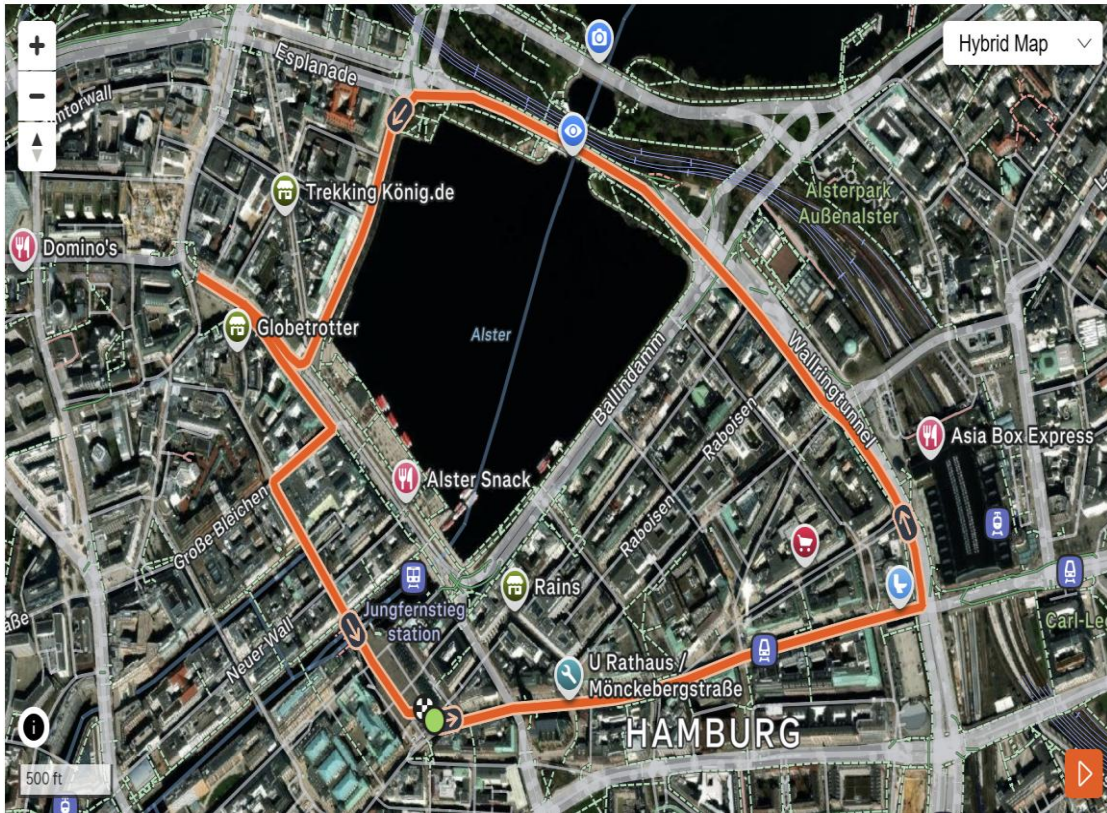
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Tactical Notes

- Single 750 m loop with a pontoon start into the Alster canal.
- Fresh water lake in city center. Typically fairly calm conditions
- July morning water mid-to-high teens Celsius. Potential for wetsuit, but has historically been above the limit.
- Short run into T1 from the Alster exit ramp; transition area directly adjacent to city hall

Bike Course Profile

Hamburg



19.8 km • 6 × 3.3 km loops

Closed urban circuit — Rathausmarkt, Alsterufer, Jungfernstieg

Elevation / Lap

27 ft / lap

Total Climb

162 ft

Avg Grade

~0-1%

Max Climb Grade

7%

Wind

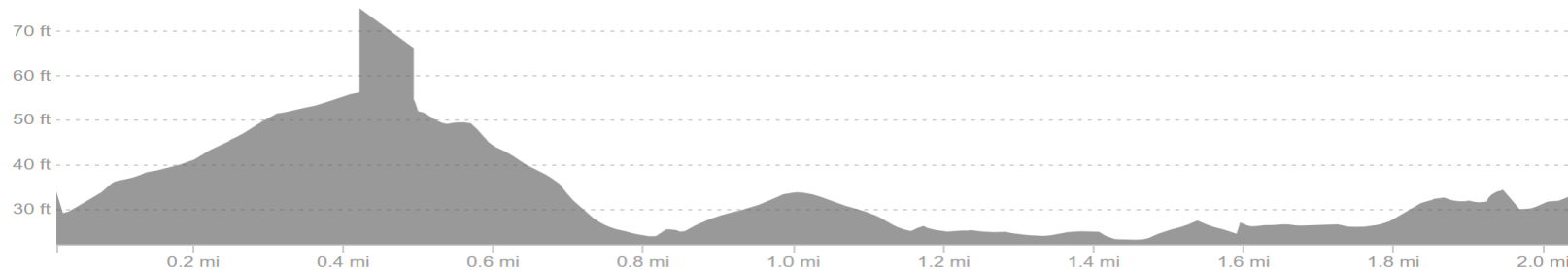
City-center setting

Loops

6 × 3.3 km

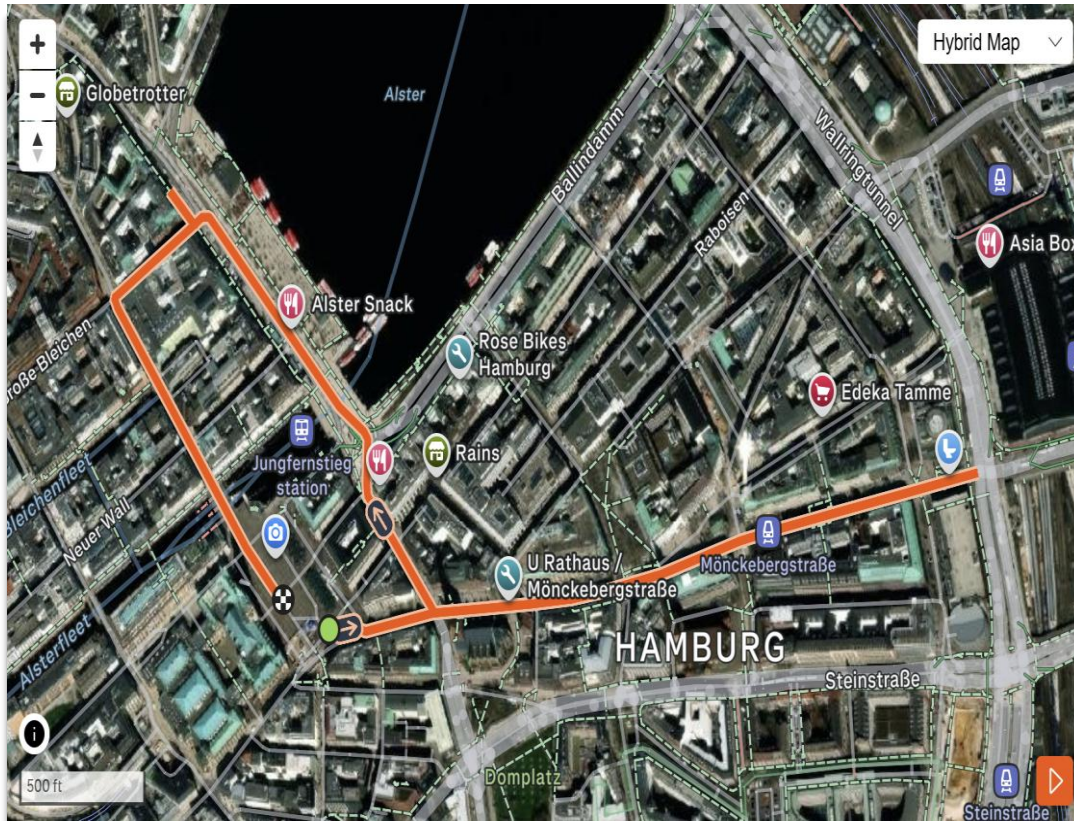
Tactical Implications

- Hamburg's city center bike represents one of the most technical WTCS bike courses. Pancake-flat throughout, but high-frequency turns reward bike-handling skills alongside power.
- Drafting packs form easily on the long straights; Hamburg historically produces large lead packs of 30+ athletes. Cobblestone sections add additional complexity

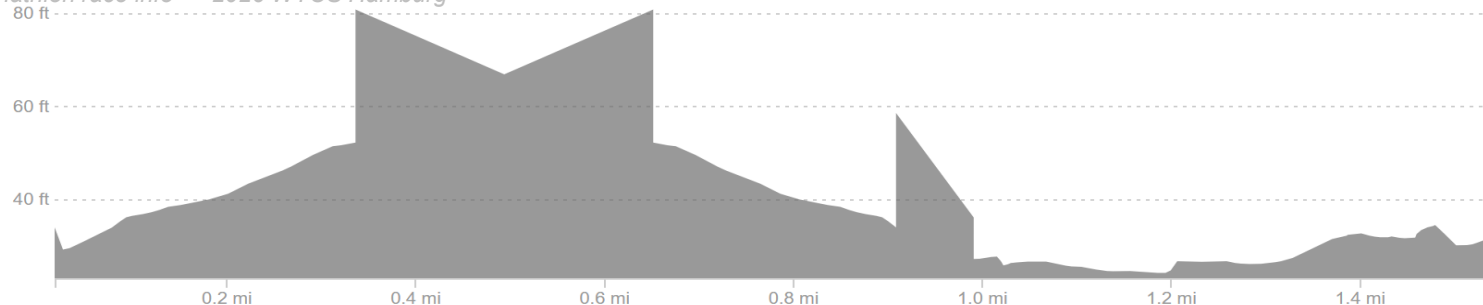


Run Course Profile

Hamburg



Source: World Triathlon race info — 2026 WTCS Hamburg



5.0 km • 2 × 2.5 km

Closed urban — Rathausmarkt, Alsterarkaden, city hall precinct

Distance
5.0 km

Lap Structure
2 × 2.5 km

Total Climb
136 ft

Avg Grade
~0-1%

Max ▲ / ▼
+ 1.8%

Heat Risk
LOW-MODERATE

Tactical Notes

- 2 × 2.5 km laps through Hamburg's historic city center offers a flat, fast, and most importantly a spectator-dense course
- Cobblestone sections add additional terrain from typically road surfaces.
- July Hamburg averages ~22 °C, manageable but warm

Travel & Arrival Load

Denver, CO → Hamburg race week



Core travel read

Denver → Hamburg requires one or two stops. Typical routes connect through London (LHR), Amsterdam (AMS), Frankfurt (FRA), or Paris (CDG) before the short connection into Hamburg (HAM). Lufthansa and United operate the best options from DEN. CEST (UTC+2) is 8 hours ahead of MDT. Hamburg Airport (HAM) is 10 km from the city center. Taxi/rideshare ~20 min, S-Bahn line S1 links the airport to the Hauptbahnhof in ~25 min. Most WTCS hotels sit within 1–2 km of the Alster.

Primary route

DEN → HAM

1–2 stops

Flight time

~14–16h

Including layover

Time zones

+8h

CEST ahead of MDT

Airport Transfer

HAM → City Center

S-Bahn S1 ~25 min or taxi ~20 min

Transfer Distance

~10 km

HAM Airport to Binnenalster

Host Hotel / Official Accommodation

- Historic WTCS host hotels cluster along the Außenalster and inner city: Radisson Blu, Fairmont Hotel Vier Jahreszeiten, and NH Hotels are common LOC choices.
- Binnenalster / Rathausmarkt area gives walking distance to transition. No shuttle needed for pre-race check-in or familiarization.
- City centre parking is limited and expensive. most athletes walk or use S-Bahn from the hotel to the venue.

Food / Grocery Options Near Hotel + Venue

- Best stock-up: REWE, Edeka, or Aldi branches within 10 min walk of the Binnenalster area; Hamburger Str / Mönckebergstraße corridor has most options.
- Quick basics: dm (pharmacy), Starbucks, Backwerk, and bakeries throughout the Rathausmarkt pedestrian zone.
- Athlete-friendly meals: Zum Alten Rathaus, Block House (Hamburg chain), and Superkitchen near the Alster area.

Environmental Risk Profile

Hamburg



RACE-DAY WEATHER (08:00–11:00 local)						AIR QUALITY (Open-Meteo archive)				
Year	Date	Air Temp	Humidity	Wind km/h	Precip mm	Year	PM2.5	EU AQI	UV Index	Risk
2018	Jul 14	21	67%	11	0.0	2018	2.5	19	—	Good
2019	Jul 06	15.0°*	87%	16	1.1	2019	6.2	14	—	Good
2020	Sep 05	13.8°*	94%	14	—	2020	4.8	17	—	Good
2021	Sep 18	15.5°*	84%	10	0.0	2021	3.8	10	—	Good
2022	Jul 09	23.4	81%	—	0.1	2022	4.7	15	—	Good
2023	Jul 15	24.4°*	54%	—	0.0	2023	6.5	35	2.8	Fair
2024	Jul 13	18.9	79%	22	0.0	2024	3.6	21	1.2	Fair
2025	Jul 12	23.6	76%	14	0.2	2025	5.7	21	1.5	Fair

HEALTH & WATER QUALITY RISK FACTORS

- Race Report Here -> [Water Quality | 2026 World Triathlon Championship Series Hamburg](#)

Weather and AQ data: Open-Meteo historical archive (archive-api.open-meteo.com). EU AQI: Good ≤20, Fair ≤40, Moderate ≤60, Poor ≤80, Very Poor >80.

Who to Watch — Hamburg Race Week



ELITE MEN

Top 10 by World Ranking

Rank	Athlete	Points
1	Vasco Vilaca	6,789
2	Miguel Hidalgo	6,703
3	Matthew Hauser	6,341
4	Henry Graf	5,726
5	David Cantero Del Campo	5,062
6	Csongor Lehmann	5,030
7	Dorian Coninx	4,545
8	Luke Willian	4,534
9	Ricardo Batista	4,458
10	Alessio Crociani	3,921

Last Year's Podium at Hamburg

1. Matthew Hauser (00:50:07)
2. Vasco Vilaca (00:50:14)
3. Alessio Crociani (00:50:36)

ELITE WOMEN

Top 10 by World Ranking

Rank	Athlete	Points
1	Beth Potter	7,399
2	Lisa Tertsch	6,914
3	Jeanne Lehair	6,295
4	Leonie Periault	6,170
5	Cassandre Beaugrand	5,591
6	Jolien Vermeylen	5,111
7	Taylor Spivey	4,879
8	Emma Lombardi	4,767
9	Diana Isakova	4,677
10	Bianca Seregni	4,659

Last Year's Podium at Hamburg

1. Leonie Periault (00:56:25)
2. Cassandre Beaugrand (00:56:29)
3. Beth Potter (00:56:32)

Rankings = current World Triathlon rankings snapshot; startlist intersect uses program_entries when available.

Top Swim Threats

Hamburg



Ranked by EWMA swim split (span-5, most recent) • Lead Pack % = within 15 s of swim leader • WTCS / Championship Finals

ELITE MEN						ELITE WOMEN					
#	Athlete	Lead Pack %	Avg Gap (s)	EWMA Swim	Races	#	Athlete	Lead Pack %	Avg Gap (s)	EWMA Swim	Races
1	Jan Diener	50%	+13.5	8:38	2	1	Cecilia Perez	0%	+33.0	9:36	2
2	Kauê Willy	50%	+21.5	8:43	2	2	Carlotta Missaglia	0%	+44.0	9:49	2
3	Gjalt Panjer	50%	+14.0	8:51	2	3	Karolina Helga Horváth	0%	+23.5	9:50	2
4	Kyle Smith	0%	+28.0	8:53	2	4	Sandra Dodet	0%	+40.5	9:50	2
5	Jannik Schaufler	0%	+19.7	9:01	3	5	Nele Dequae	0%	+50.0	9:56	2
6	Brayden Mercer	100%	+10.5	9:01	2	6	Margot Garabedian	0%	+25.5	10:01	2
7	James Edgar	0%	+27.5	9:10	4	7	Julia Bröcker	0%	+34.7	10:09	3

U23 MEN						U23 WOMEN					
#	Athlete	Lead Pack %	Avg Gap (s)	EWMA Swim	Races	#	Athlete	Lead Pack %	Avg Gap (s)	EWMA Swim	Races
1	Pavel Sorokin	100%	+11.0	17:41	2	1	Tara Sosinski	100%	+6.5	18:45	2
2	Jules Rethoret	100%	+11.5	17:41	2	2	Angelica Prestia	100%	+9.5	18:48	2
3	Esteban Basanta Fouz	50%	+14.0	17:43	2	3	Sophia Howell	100%	+1.0	19:03	3
4	Mitch Kolkman	100%	+10.8	17:50	4	4	Selina Klamt	50%	+19.5	19:04	2
5	Nicolas Probert Vargas	50%	+27.5	17:54	2	5	Jessica Fullagar	50%	+20.0	19:05	2
6	Sebastian Wernersen	25%	+22.0	18:00	4	6	Ilona Hadhoum	100%	+4.0	19:13	2
7	Alessio Crociani	100%	+0.0	18:06	2	7	Heidi Juránková	0%	+45.5	19:19	2

USA World Rankings Snapshot

Hamburg



As of 2026-07-06 • World Triathlon Rankings • Heading into Hamburg • Race tier: WTCS

Max Points Available at this Race

Win 1,000 pts	Podium 855 pts	Top 5 732 pts	Top 10 495 pts	Top 15 335 pts	Top 20 227 pts
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ELITE MEN

Rank	Athlete	Points	Curr	Prev
16	John Reed	3,147	6	6
26	Reese Vannerson	2,628	6	6
49	Braxton Legg	2,088	6	6
55	Morgan Pearson	2,026	4	5
64	Sullivan Middaugh	1,910	6	6
67	Chase McQueen	1,839	6	6
68	Keller Norland	1,835	6	6
75	Seth Rider	1,748	6	5
83	Blake Bullard	1,696	6	6
100	Darr Smith	1,544	6	6
117	Porter Middaugh	1,335	6	4
119	Zach Leachman	1,330	6	1
166	Carter Stuhlmacher	1,044	6	3
309	Luke Anthony	520	2	6
333	Matthew McGoey	466	6	5

ELITE WOMEN

Rank	Athlete	Points	Curr	Prev
7	Taylor Spivey	4,879	6	3
26	Gwen Jorgensen	2,674	6	6
28	Gina Sereno	2,667	5	6
35	Danielle Orie	2,432	6	6
47	Erika Ackerlund	2,055	5	6
53	Naomi Ruff	1,988	6	6
58	Kelly Wetteland	1,858	6	4
84	Kirsten Kasper	1,551	5	2
85	Eleanor Beveridge	1,544	6	6
94	Joy Gill	1,479	6	3
102	Michelle Magnani	1,365	6	6
115	Rachel Werking	1,253	6	1
142	Ellison Wolfe	1,085	6	2
151	Katie McCune	1,042	6	1
160	Taylor Knibb	1,003	2	1

Olympic Qualification Pace

Hamburg



Paris 2024 OQR cutoff (30th): 3,626.69 pts ÷ 12-event cap = 302 pts / event on pace
An athlete who averages ~302 pts per event across their best 12 events is on pace to clear the LA 2028 qualifying threshold.

FINISH POSITION AT EACH TIER THAT HITS 302-PT PACE

Tier	Win pts	Last on-pace pos	First off-pace pos (pts)	Verdict
World Championship Series	1000	#16	#17 (287 pts)	Top 16
World Cup	500	#7	#8 (290 pts)	Top 7

USA MEN								USA WOMEN							
Rk	Athlete	Total	Races	On	Off	Pace	Proj @12	Rk	Athlete	Total	Races	On	Off	Pace	Proj @12
8	Reese Vannerson	980	2	2	0	490	5,884	9	Danielle Orie	1,146	4	2	2	286	3,439
16	Chase McQueen	764	3	1	2	254	3,059	12	Taylor Knibb	921	2	2	0	460	5,526
21	Sullivan Middaugh	677	2	1	1	338	4,063	18	Gina Sereno	704	2	1	1	352	4,226
27	John Reed	579	1	1	0	579	6,953	26	Taylor Spivey	579	1	1	0	579	6,953
30	Braxton Legg	564	2	1	1	282	3,384	31	Eleanor Beveridge	519	2	0	2	259	3,114
33	Seth Rider	534	2	1	1	267	3,209	36	Gwen Jorgensen	495	1	1	0	495	5,949
35	Keller Norland	519	1	1	0	519	6,234	39	Kelly Wetteland	462	1	1	0	462	5,550
41	Blake Bullard	436	2	0	2	218	2,621	40	Naomi Ruff	460	1	1	0	460	5,526
77	Zach Leachman	212	1	0	1	212	2,545	86	Rachel Werking	181	1	0	1	181	2,178
87	Cole Jamieson	169	1	0	1	169	2,038	90	Joy Gill	167	1	0	1	167	2,014
103	Porter Middaugh	115	1	0	1	115	1,380	103	Kirsten Kasper	112	1	0	1	112	1,353

On = events scoring ≥ pace. Off = events < pace. Pace = total breakdown pts ÷ events raced. Proj @12 = current total + (avg pts × remaining events to cap); green if ≥ Paris cutoff. Cutoff estimate: Paris 2024 = 3,627 pts / 12 events = 302 pts/event. Tier positions use a geometric 0.925 points model. Source: World Triathlon rankings database.

USA Olympic Qualification Snapshot

Hamburg



As of 2026-07-06 • Olympic Qualification Rankings (LA 2028 cycle) • Heading into Hamburg

ELITE MEN					ELITE WOMEN				
Rank	Athlete	Points	Second	First	Rank	Athlete	Points	Second	First
8	Reese Vannerson	980	0	2	9	Danielle Orie	1,146	0	4
16	Chase McQueen	764	0	3	12	Taylor Knibb	921	0	2
21	Sullivan Middaugh	677	0	2	18	Gina Sereno	704	0	2
27	John Reed	579	0	1	26	Taylor Spivey	579	0	1
30	Braxton Legg	564	0	2	31	Eleanor Beveridge	519	0	2
33	Seth Rider	534	0	2	36	Gwen Jorgensen	495	0	1
35	Keller Norland	519	0	1	39	Kelly Wetteland	462	0	1
41	Blake Bullard	436	0	2	40	Naomi Ruff	460	0	1
77	Zach Leachman	212	0	1	86	Rachel Werking	181	0	1
87	Cole Jamieson	169	0	1	90	Joy Gill	167	0	1
103	Porter Middaugh	115	0	1	103	Kirsten Kasper	112	0	1
125	Carter Stuhlmacher	71	0	1	115	Michelle Magnani	71	0	1

Rankings: latest weekly snapshot. Second / First = events counted in the second vs first scoring period of the LA 2028 OQR cycle. Because the cycle is still in its first year, most results appear under First. Source: World Triathlon rankings database.

ELITE MEN



Race Overview — Elite Men

Hamburg



RACE HISTORY AT HAMBURG

The Elite Men race at Hamburg has been held 8 times in this analysis window (2018–2025). Hayden Wilde and Matt Hauser have both won this race twice with Hauser the back to back defending champion.

3 KEYS TO SUCCESS

1

With the swim typically being non-wetsuit, there is a potential for a split to occur. On average around 20-25 athletes come out of the water within 15 of the lead. Overall, this swim is classified as one that's hard to win the race in, but you can certainly lose it here.

2

The Hamburg bike course is known for being highly technical and surgy. The yo-yo nature of the course does allow for motivated smaller packs to catch larger groups that are bogged down by each other. In 5 of the last 8 races, more riders joined the front and when this happens the front group is typically around ~30 athletes.

3

Exceptional run speed is required to win here especially in the last two years where a 13:40 and 13:50 was needed to take Gold. Even the average 5K is quick here with the previous two years right at the 15 min benchmark.

Results Summary — Elite Men

Hamburg



Year	Date	Category	Field	DNF/LAP/ DNS	Winner	Winner Time	Air Temp	Water Temp	Swim km	Bike km	Run km
2018	Jul 14	World Championship Series	56	9	Mola	53:24	21	20.9	0.75	21.0	5.0
2019	Jul 06	World Championship Series *	56	9	Birtwhistle	55:09	15.0°*	—	—	—	—
2020	Sep 05	World Championships, World Championship Series	65	7	Luis	49:13	13.8°*	—	—	—	—
2021	Sep 18	World Championship Series	52	4	Hellwig	53:08	15.5°*	—	—	—	—
2022	Jul 09	World Championship Series *	57	3	Wilde	53:10	23.4	20.8	0.75	21.0	5.0
2023	Jul 15	World Championships, World Championship Series, Age-Group Event *	60	1	Wilde	19:26	24.4°*	—	0.30	7.5	1.75
2024	Jul 13	World Championship Series *	57	7	Hauser	50:03	18.9	21.3	0.75	19.8	4.9
2025	Jul 12	World Championship Series *	55	6	Hauser	50:07	23.6	21.1	0.75	19.8	5.0

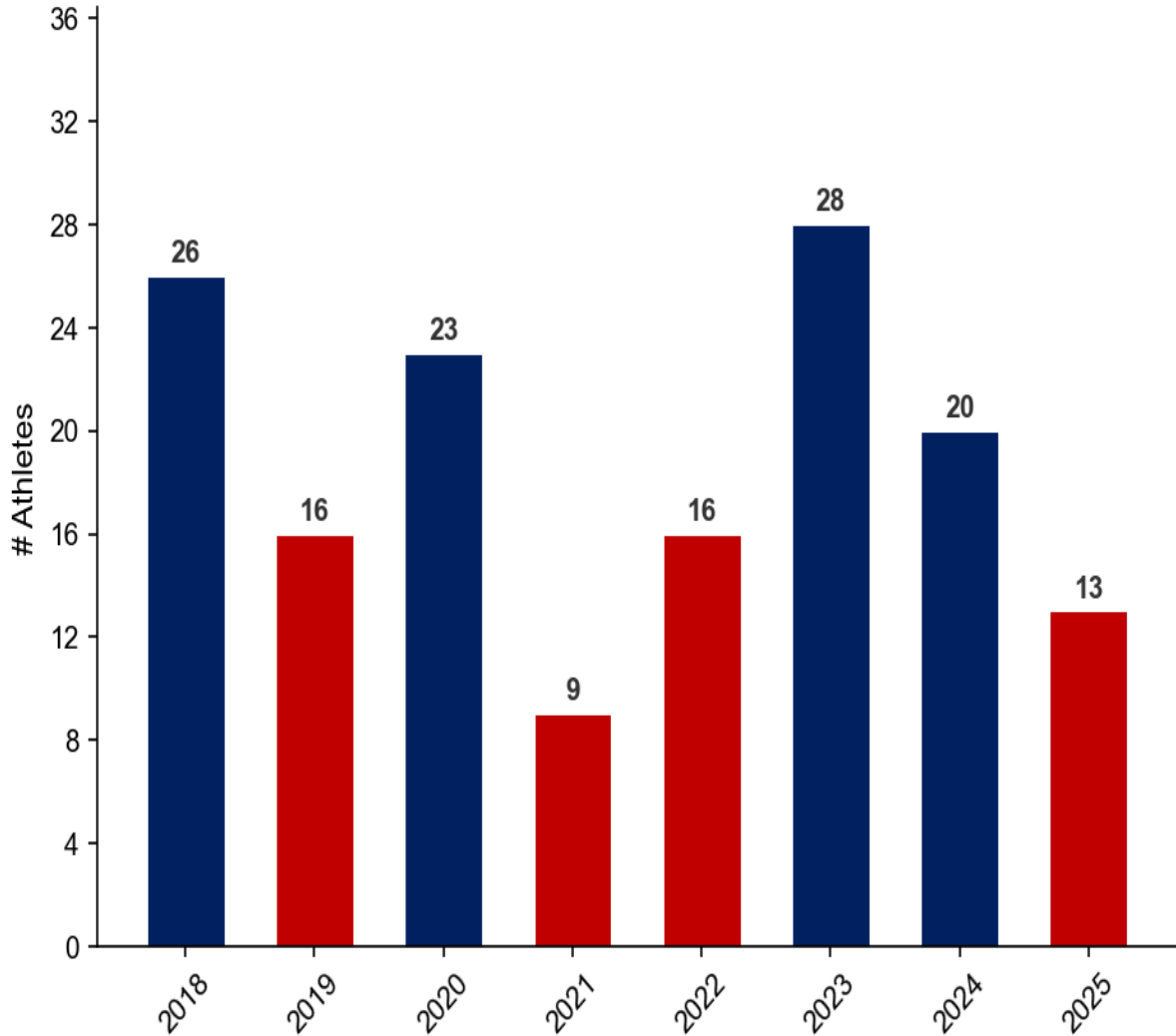
* Course distance changed vs. prior year

Swim Exit Dynamics — Elite Men

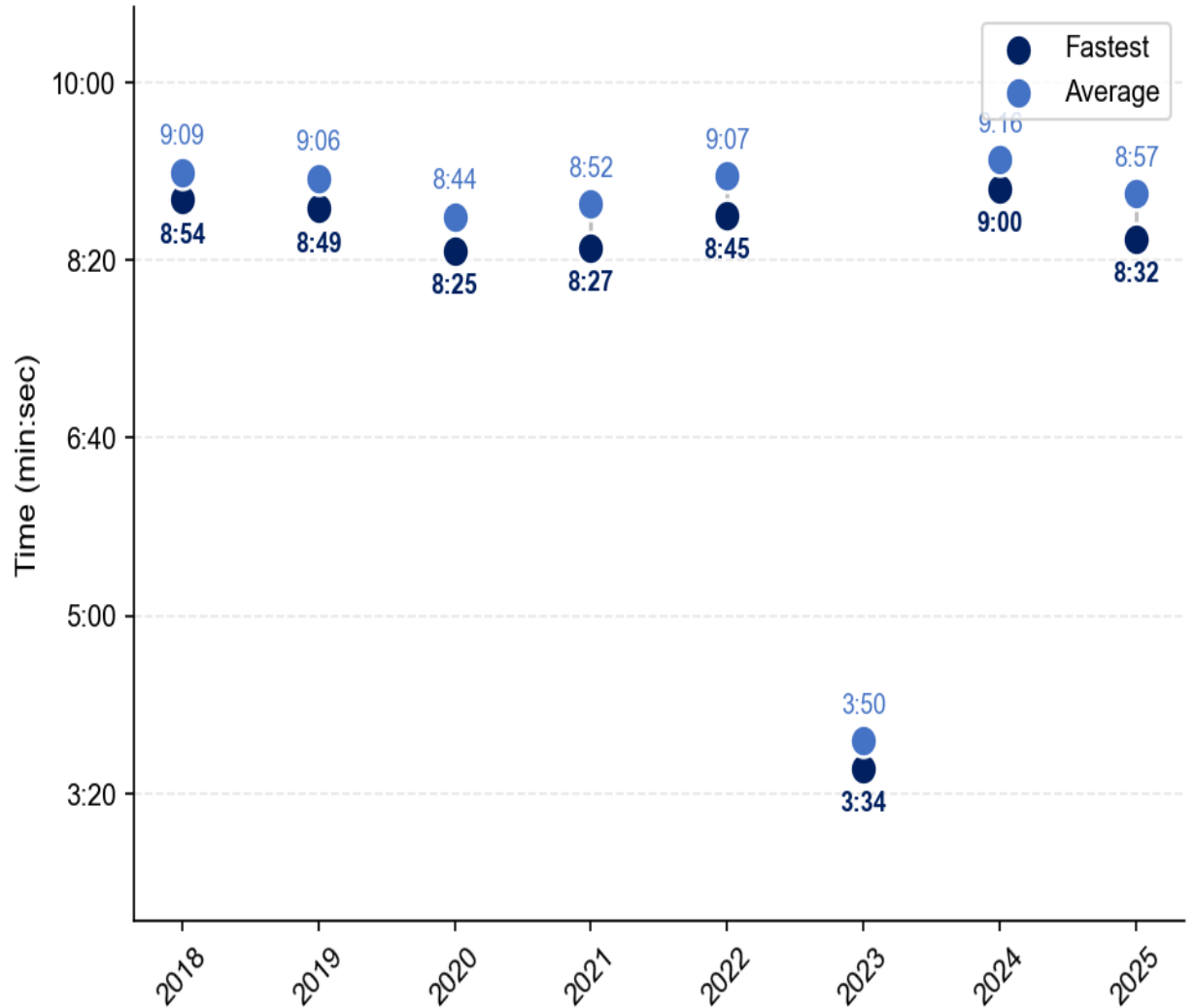
Hamburg



Athletes Within 15 sec of Swim Leader



Fastest vs. Average Swim Time

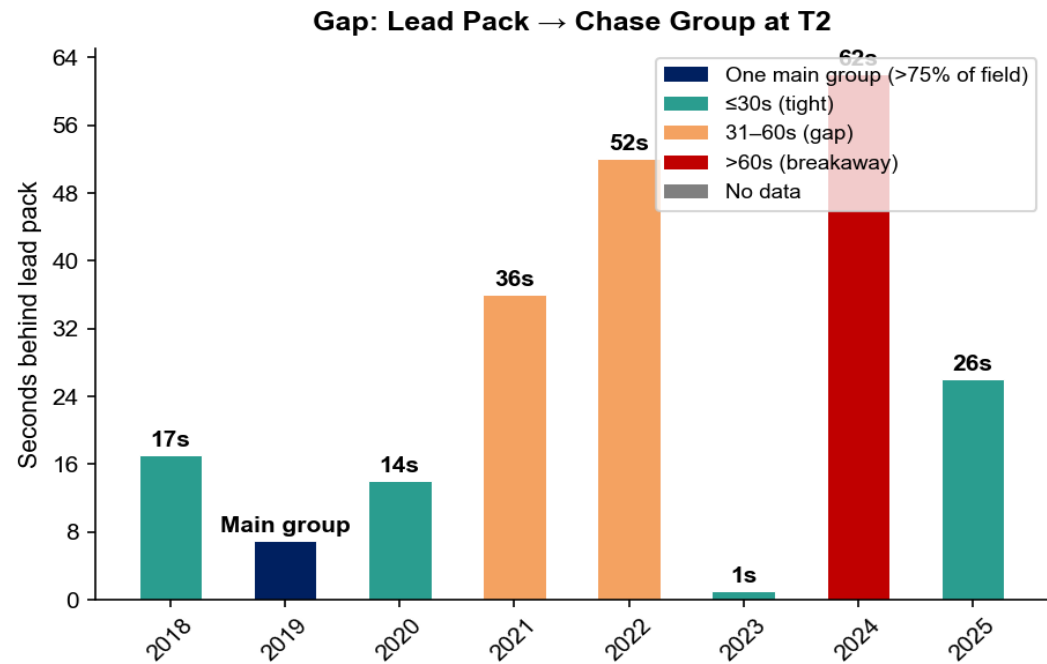
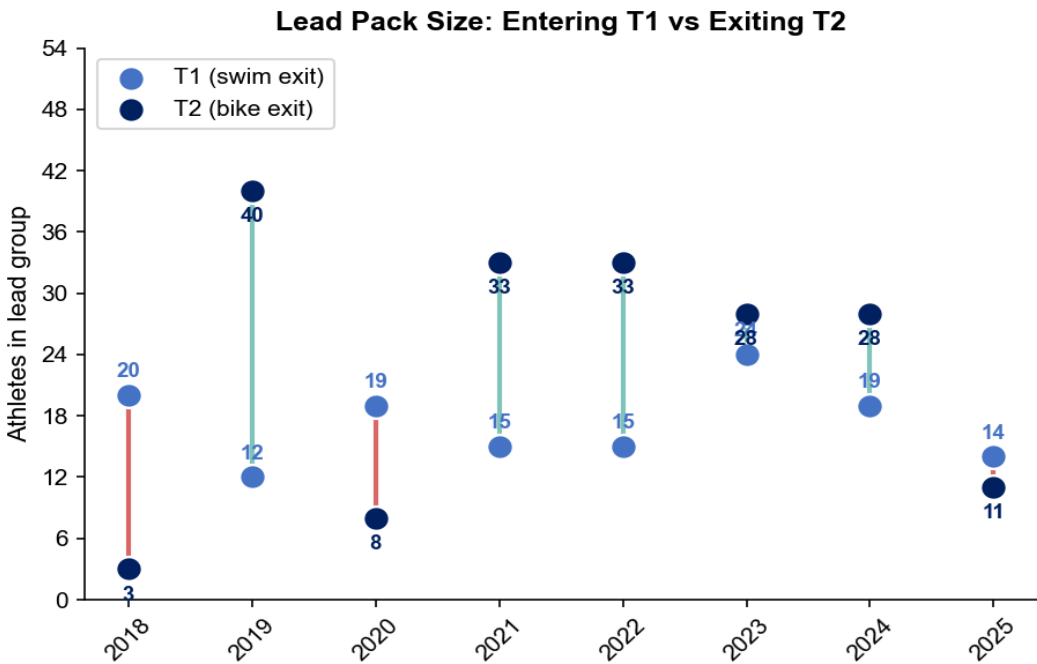


Bike Pack Evolution — Elite Men

Hamburg



Lead pack = athletes within 15 sec of leader. T1 = lead group entering the bike; T2 = lead group exiting the bike.



Year	Lead Size @ T1	Lead Size @ T2	Change	Chase Group @ T2	Chase Gap*
2018	20	3	-17	37	17s
2019	12	40	+28	2	28s
2020	19	8	-11	34	14s
2021	15	33	+18	9	36s
2022	15	33	+18	13	52s
2023	24	28	+4	23	1s
2024	19	28	+9	16	62s
2025	14	11	-3	20	26s

Lead pack = athletes within 15 sec of leader. Chase group = athletes within 15 sec of the first athlete outside the lead pack. * Chase Gap = seconds from the last lead-pack athlete to the first chase-group athlete at T2.

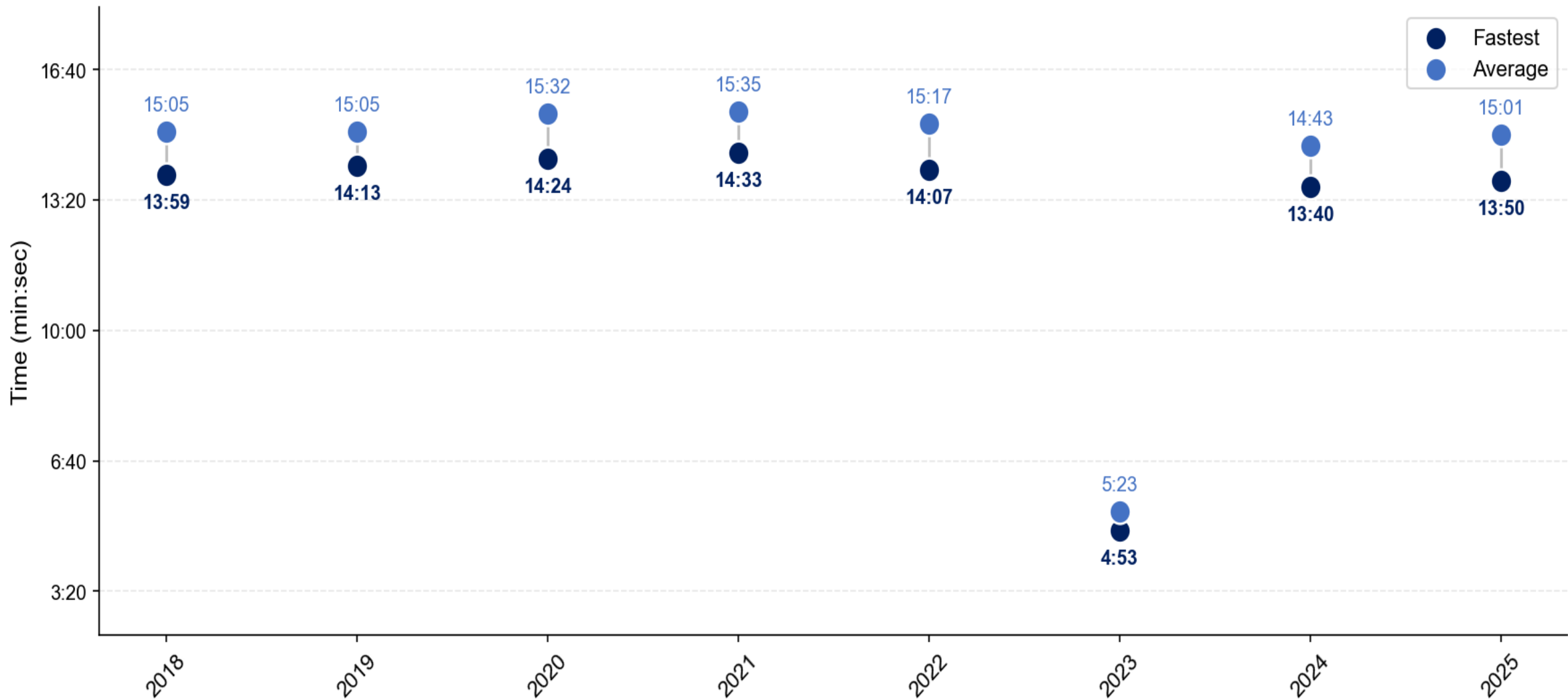
Dropouts: 2018: 8 DNF/LAP (1 in lead group at t1) | 2019: 8 DNF/LAP | 2020: 6 DNF/LAP (1 in lead group at t1) | 2021: 2 DNF/LAP | 2022: 1 DNF/LAP | 2024: 5 DNF/LAP (1 in lead group at t1) | 2025: 6 DNF/LAP (1 in lead group at t1)

Run Split Analysis — Elite Men

Hamburg



Fastest vs. Average Run Split by Year

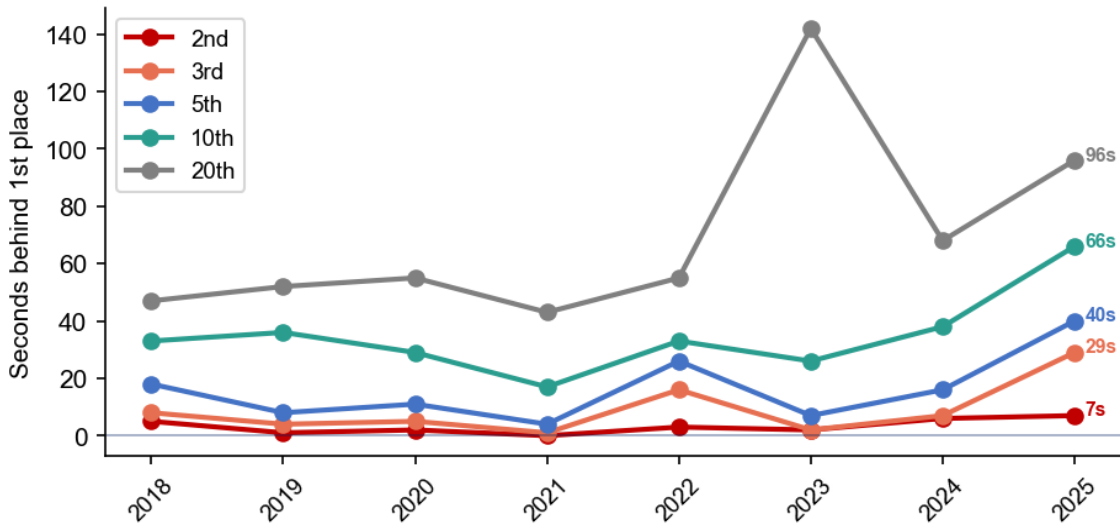


Key Position Times — Elite Men

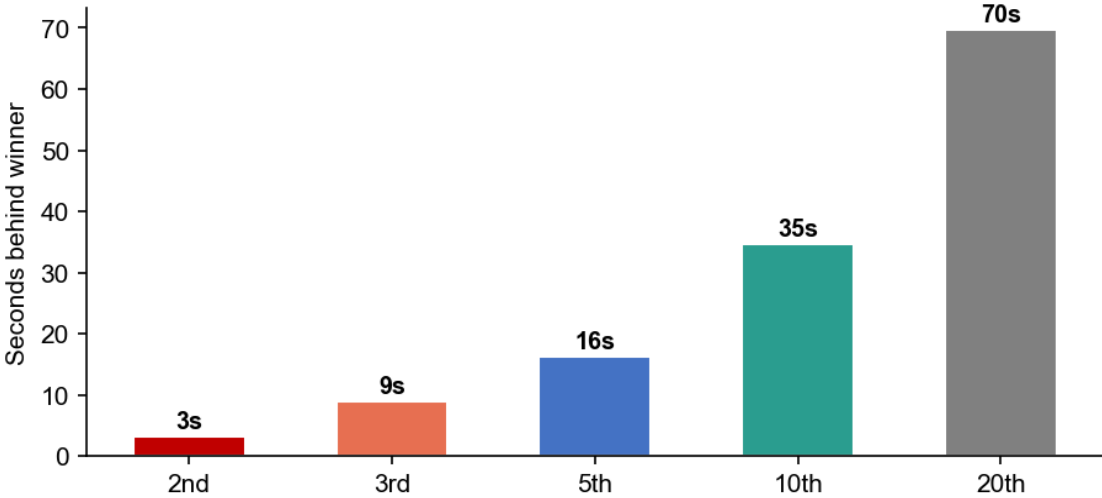
Hamburg



Gap from Winner (seconds)



Avg Gap from 1st Place (all years)



Year	1st	2nd	3rd	5th	10th	20th
2018	53:24	53:29 +5s	53:32 +8s	53:42 +18s	53:57 +33s	54:11 +47s
2019	55:09	55:10 +1s	55:13 +4s	55:17 +8s	55:45 +36s	56:01 +52s
2020	49:13	49:15 +2s	49:18 +5s	49:24 +11s	49:42 +29s	50:08 +55s
2021	53:08	53:08	53:09 +1s	53:12 +4s	53:25 +17s	53:51 +43s
2022	53:10	53:13 +3s	53:26 +16s	53:36 +26s	53:43 +33s	54:05 +55s
2023	19:26	19:28 +2s	19:28 +2s	19:33 +7s	19:52 +26s	21:48 +142s
2024	50:03	50:09 +6s	50:10 +7s	50:19 +16s	50:41 +38s	51:11 +68s
2025	50:07	50:14 +7s	50:36 +29s	50:47 +40s	51:13 +66s	51:43 +96s

Race Conditions — Elite Men

Hamburg



Year	Date	Air Temp	Humidity	Wind (km/h)	Precip (mm)	Water Temp	Wetsuit	Conditions
2018	Jul 14	21	67%	11	0.0	20.9	forbidden	—
2019	Jul 06	15.0**	87%	16	1.1	—	—	—
2020	Sep 05	13.8**	94%	14	—	—	—	—
2021	Sep 18	15.5**	84%	10	0.0	—	—	—
2022	Jul 09	23.4	81%	—	0.1	20.8	forbidden	—
2023	Jul 15	24.4**	54%	—	0.0	—	forbidden	—
2024	Jul 13	18.9	79%	22	0.0	21.3	forbidden	—
2025	Jul 12	23.6	76%	14	0.2	21.1	forbidden	—

Air temp, humidity, wind, and precipitation sourced from Open-Meteo historical archive (race window 08:00–11:00 local) where WT API data is absent. Values marked * are Open-Meteo estimates. Water temp and wetsuit ruling are from the World Triathlon API per program.

ELITE WOMEN



Race Overview — Elite Women

Hamburg



RACE HISTORY AT HAMBURG

The Elite Women race at Hamburg has been held 8 times in this analysis window (2018–2025). Cassandre Beaugrand is the most successful athlete in this span with 3 victories. However, last year Leonie Periault took the win after nearly the entire group came together on the bike.

3 KEYS TO SUCCESS

1

Disregarding the shortened 2023 course, the swim here has typically significantly shortened the field. 2025 had a very small breakaway of 4 athletes emerge, and '21 and '22 both had only 15 athletes within the 15 second benchmark. Expect for strong swimmer to push the pace with the non wetsuit conditions.

2

Historically, the women's race has seen a bike group be able to put a gap on the chasers. While typically this lead group isn't a small pack of less than 10, it also usually remains less than 30 riders. That changed last year though when the front pack ballooned to 45. The dynamic will depend on if the lead remains small enough to cooperate and hold speed through the turns.

3

Run time have continued to progress on the Hamburg course. With the leaders running 15:20 and 15:28 respectively over the past two years. Gap last year to the podium were also quite tight with 2nd and 3rd only 4 and 7 seconds back from the winner.

Results Summary — Elite Women

Hamburg



Year	Date	Category	Field	DNF/L AP/DNS	Winner	Winner Time	Air Temp	Water Temp	Swim km	Bike km	Run km
2018	Jul 14	World Championship Series	56	8	Beaugrand	58:06	20.9	21.3	0.75	21.0	5.0
2019	Jul 06	World Championship Series *	53	6	Stanford	59:24	15.0°*	—	—	—	—
2020	Sep 05	World Championships, World Championship Series	65	5	Taylor-Brown	54:16	13.8°*	—	—	—	—
2021	Sep 18	World Championship Series	46	—	Lindemann	58:17	15.5°*	—	—	—	—
2022	Jul 09	World Championship Series *	56	3	Duffy	58:37	22.5	20.4	0.75	21.0	5.0
2023	Jul 15	World Championships, World Championship Series, Age-Group Event *	57	1	Beaugrand	21:35	24.4°*	—	0.30	7.5	1.75
2024	Jul 13	World Championship Series *	55	4	Beaugrand	55:19	19.2	21.2	0.75	19.8	4.9
2025	Jul 12	World Championship Series *	55	3	Periault	56:25	21.3	21.	0.75	19.8	5.0

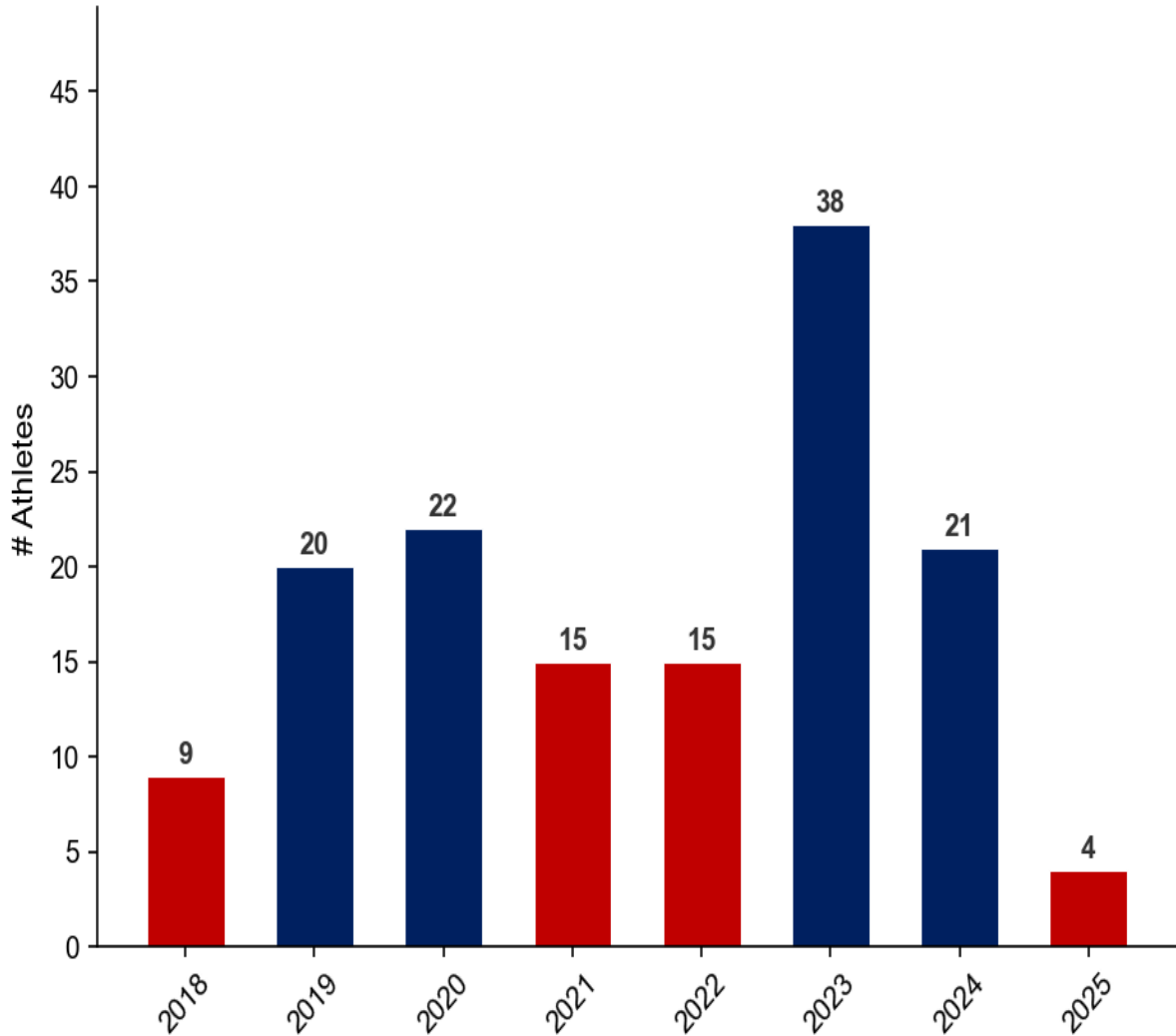
* Course distance changed vs. prior year

Swim Exit Dynamics — Elite Women

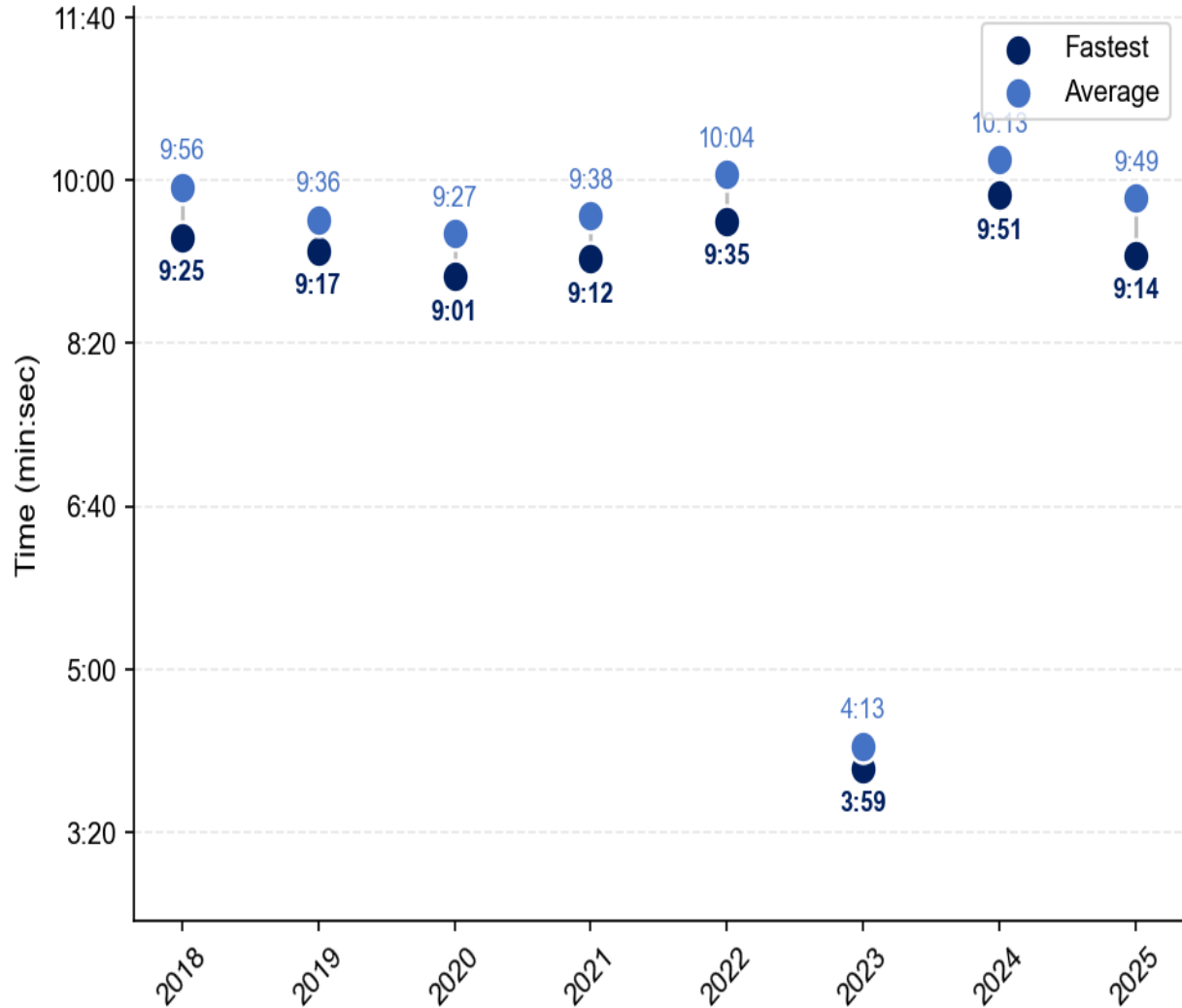
Hamburg



Athletes Within 15 sec of Swim Leader



Fastest vs. Average Swim Time



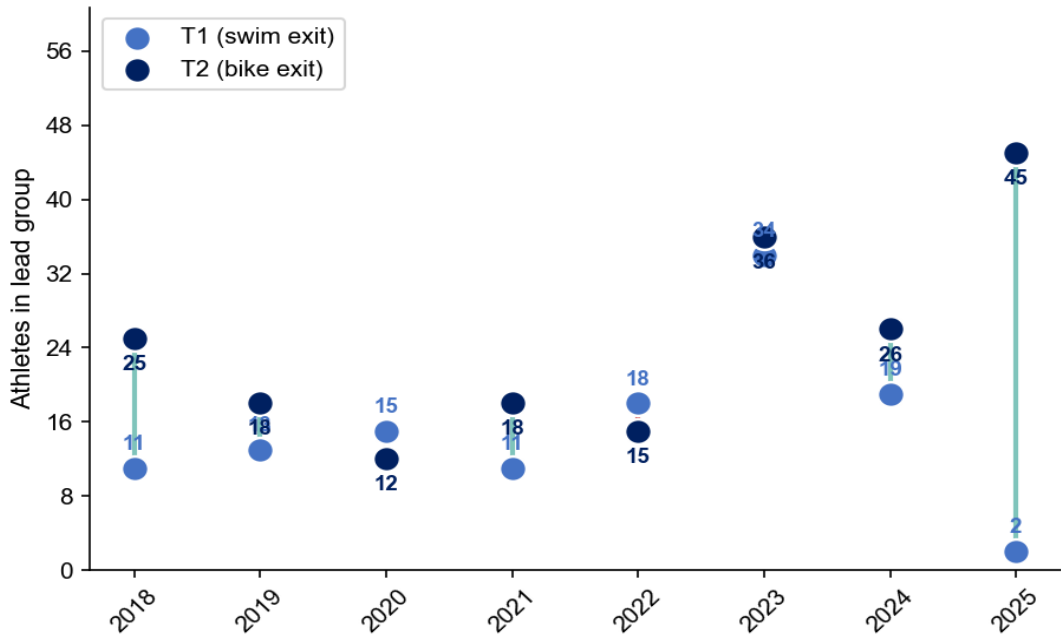
Bike Pack Evolution — Elite Women

Hamburg

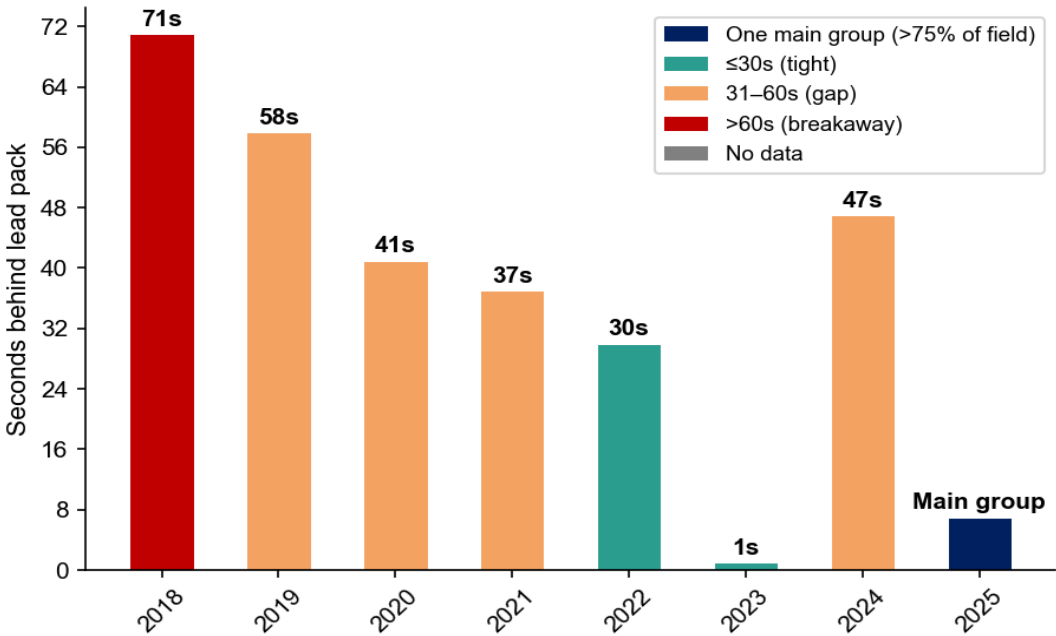


Lead pack = athletes within 15 sec of leader. T1 = lead group entering the bike; T2 = lead group exiting the bike.

Lead Pack Size: Entering T1 vs Exiting T2



Gap: Lead Pack → Chase Group at T2



Year	Lead Size @ T1	Lead Size @ T2	Change	Chase Group @ T2	Chase Gap*
2018	11	25	+14	12	71s
2019	13	18	+5	7	58s
2020	15	12	-3	5	41s
2021	11	18	+7	13	37s
2022	18	15	-3	27	30s
2023	34	36	+2	11	1s
2024	19	26	+7	10	47s
2025	2	45	+43	7	82s

Lead pack = athletes within 15 sec of leader. Chase group = athletes within 15 sec of the first athlete outside the lead pack. * Chase Gap = seconds from the last lead-pack athlete to the first chase-group athlete at T2.

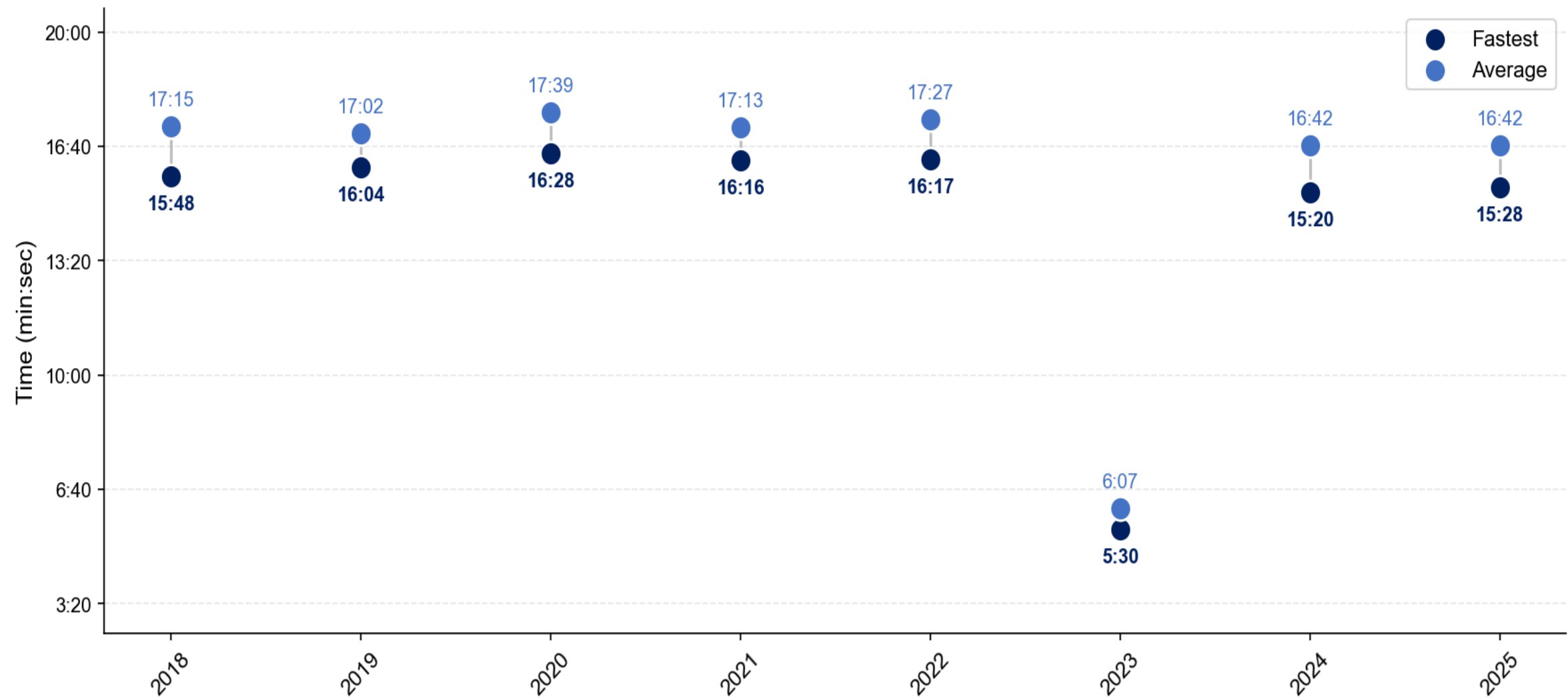
Dropouts: 2018: 6 DNF/LAP | 2019: 5 DNF/LAP | 2020: 5 DNF/LAP | 2022: 2 DNF/LAP (1 in lead group at t1) | 2024: 3 DNF/LAP (1 in lead group at t1) | 2025: 3 DNF/LAP

Run Split Analysis — Elite Women

Hamburg

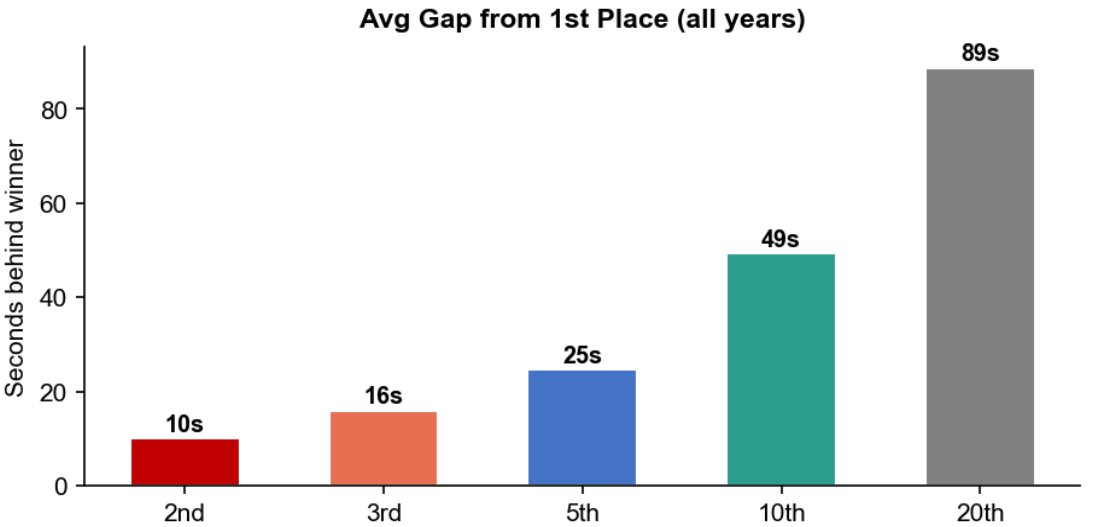
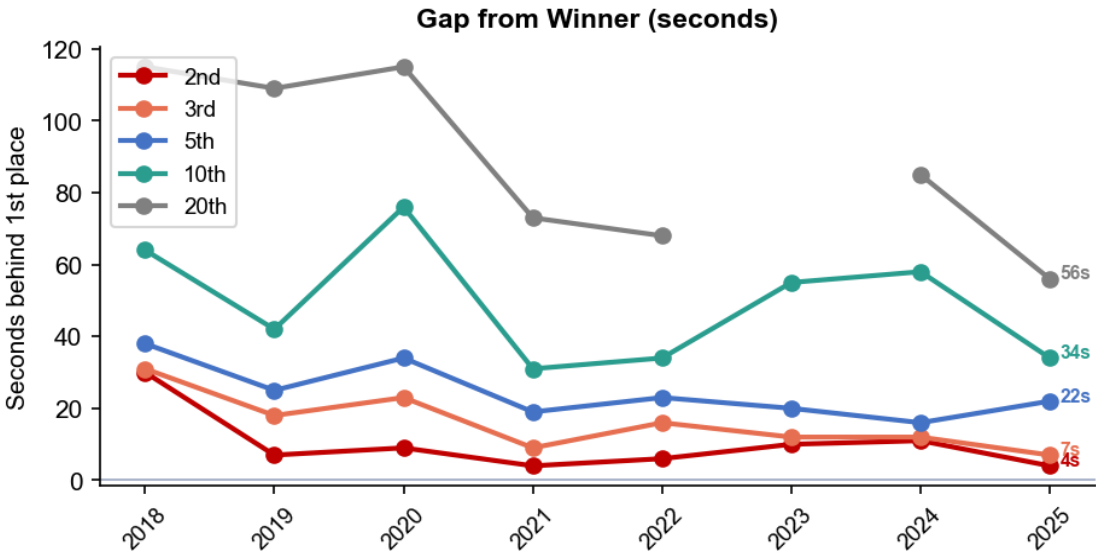


Fastest vs. Average Run Split by Year



Key Position Times — Elite Women

Hamburg



Year	1st	2nd	3rd	5th	10th	20th
2018	58:06	58:36 +30s	58:37 +31s	58:44 +38s	59:10 +64s	1:00:01 +115s
2019	59:24	59:31 +7s	59:42 +18s	59:49 +25s	1:00:06 +42s	1:01:13 +109s
2020	54:16	54:25 +9s	54:39 +23s	54:50 +34s	55:32 +76s	56:11 +115s
2021	58:17	58:21 +4s	58:26 +9s	58:36 +19s	58:48 +31s	59:30 +73s
2022	58:37	58:43 +6s	58:53 +16s	59:00 +23s	59:11 +34s	59:45 +68s
2023	21:35	21:45 +10s	21:47 +12s	21:55 +20s	22:30 +55s	—
2024	55:19	55:30 +11s	55:31 +12s	55:35 +16s	56:17 +58s	56:44 +85s
2025	56:25	56:29 +4s	56:32 +7s	56:47 +22s	56:59 +34s	57:21 +56s

Race Conditions — Elite Women

Hamburg



Year	Date	Air Temp	Humidity	Wind (km/h)	Precip (mm)	Water Temp	Wetsuit	Conditions
2018	Jul 14	20.9	67%	11	0.0	21.3	forbidden	—
2019	Jul 06	15.0**	87%	16	1.1	—	—	—
2020	Sep 05	13.8**	94%	14	—	—	—	—
2021	Sep 18	15.5**	84%	10	0.0	—	—	—
2022	Jul 09	22.5	81%	—	0.1	20.4	forbidden	—
2023	Jul 15	24.4**	54%	—	0.0	—	forbidden	—
2024	Jul 13	19.2	79%	22	0.0	21.2	forbidden	—
2025	Jul 12	21.3	76%	14	0.2	21.	forbidden	—

Air temp, humidity, wind, and precipitation sourced from Open-Meteo historical archive (race window 08:00–11:00 local) where WT API data is absent. Values marked * are Open-Meteo estimates. Water temp and wetsuit ruling are from the World Triathlon API per program.